FAMILY COURSE SYLLABUS

Competitive VEX IQ: Level Up

2026-27 | Mary R. Stauffer Middle School | Teacher: Ms. Lang

OUR COURSE PROMISE Build boldly. Compete kindly. Students learn engineering by designing, testing, documenting, and improving together.

Welcome, families

This year-long elective uses the VEX IQ Robotics Competition game Level Up as an authentic engineering challenge. The course follows the official VEX Robotics competition pathway administered by the Global Robotics & Science Foundation. Students will build mechanisms, program robots, practice driving, analyze data, communicate design decisions, and maintain an engineering notebook. Competition gives the work a real audience; thoughtful evidence makes the learning visible.

A typical week

Day	Time	Focus	What students do
MONDAY	47-52 min	**Learn + launch**	Direct instruction, teacher demonstrations, safety, and the week's engineering target.
WEDNESDAY	80-81 min	**VEX IQ STEM Lab**	Teams build, code, test, and document an official VEX IQ STEM Lab activity.
FRIDAY	80-81 min	**Level Up competition lab**	Teams improve the competition robot, practice skills, study strategy, scrimmage, and reflect.

Every class ends with notebook evidence, one documented next step, charged batteries, labeled work, returned parts, and a clean field.

Six-person competition teams

Each team includes three lead-and-backup pairs. A backup is fully trained and ready to assume the same core work.

- Two drivers: lead driver and backup driver
- Two programmers: lead programmer and backup programmer
- Two strategists: lead strategist and backup strategist

All six students also share construction, inspection, testing, cleanup, parts stewardship, and engineering-notebook evidence. Returning experience is distributed across teams; lead roles are based on readiness and follow-through, not seniority. At an event, Ms. Lang assigns a trained Driver 1, Driver 2, and Loader for each match; those event positions are separate from the classroom specialty-pair titles.

How students are graded

Grades reward the quality of the learning process, individual contribution, technical growth, and communication. Tournament placement and trophies never determine a student's grade.

Weight	Category	Examples of evidence
30%	**Engineering process**	Notebook entries, sketches, iterations, and evidence
25%	**Technical performance**	Build quality, code, driving, reliability, and growth
20%	**Collaboration**	Role follow-through, communication, safety, and stewardship
15%	**Competition readiness**	Rules, inspection, scouting, interviews, and match conduct
10%	**Reflection + communication**	Design reviews, portfolios, demonstrations, and handoff

Engineering notebook expectations

Teams use the engineering notebook to show the engineering design cycle: identify a problem, brainstorm alternatives, select with evidence, build or program, test, interpret, and iterate. Entries should be dated, readable, specific, and connected to student work.

- Identify the problem, goal, or constraint.
- Include sketches, code notes, measurements, photos, or trial data when useful.
- Explain what the evidence means, not only what happened.
- Record individual contributions and the team's next test.
- Keep failed ideas and raw data; duplicate template pages for each meaningful session or design cycle.

Major learning milestones

Window	Milestone	Family-facing evidence
Aug-Sep	**Launch the Lab**	Safety, team charter, rules, and baseline data
Sep-Oct	**Mechanisms That Score**	Prototype tests, design review, and robot v1
Nov-Dec	**Drive, Code, Compete**	Driving, autonomous, mock event, and portfolio
Jan-Feb	**Rebuild From Evidence**	Failure analysis, robot v2, A/B test, and benchmark
Feb-Apr	**Peak Competition**	Scouting, inspection, judging, and event reflection
Apr-Jun	**Legacy + Showcase**	Student teaching, portfolio, showcase, and handoff

Competition and event commitments

Competition dates may occur outside the regular school day. Ms. Lang will publish confirmed events, transportation information, permission requirements, arrival times, and packing expectations in Canvas. Participation expectations will be communicated in advance, and students should notify the teacher promptly about conflicts.

- Students represent Stauffer through safe, respectful, and student-centered conduct.
- Teams keep robots legal, inspected, charged, labeled, and event-ready.
- Students should be able to explain their own robot, code, strategy, and design process.
- The current official game manual and updates are always the authority.

LEVEL UP BASELINE Course materials were checked against Game Manual Version 1.1 (August 6, 2026): 60-second Teamwork and Skills matches, an 11 x 20 x 15-inch starting Robot, no more than six Smart Motors, and Driver 1/Driver 2/Loader procedures. Teams recheck the live manual before events.

How families can help

- Ask: What did your team test? What evidence changed your mind? What will you try next?
- Help students arrive prepared for announced events and meet permission deadlines.
- Celebrate persistence, documentation, teamwork, and technical growth - not only scores.
- Encourage students to solve team problems directly and respectfully before adults take over.
- Contact Ms. Lang when health, transportation, accessibility, or family circumstances affect participation.

Safety, care, and academic integrity

Students must follow lab safety directions, use batteries and tools responsibly, keep aisles and fields clear, and return shared parts. The robot, code, notebook, and judged explanations must represent student work. Adults and digital tools may support learning, but may not complete competition work for students.

Communication

Canvas is the primary home for announcements, assignments, resources, confirmed event details, and due dates. Families should use the school's normal communication channels to contact Ms. Lang.

FAMILY CHECK-IN Please review this syllabus with your student, confirm that Canvas notifications are working, and save confirmed competition dates when they are announced.

Important links

Curriculum hub: https://spartandesign.github.io/spartandesign-viqrc/

The curriculum hub maintains links to VEX IQ STEM Labs, the current Level Up manual, official team resources, judging guidance, and event information.

The current game manual and official updates supersede printed summaries.